COURSE BOOKLET

B.Tech. 1st Year

Computer Science and Information Technology (CSIT)



CURRICULUM STRUCTURE & SYLLABUS

Effective from the Session: 2026-27

UNIVERSITY VISION

To evolve as a globally recognized multidisciplinary university committed to skill-based education, research, innovation, and ethical values, nurturing holistic individuals to serve society for sustainable development.

UNIVERSITY MISSION

- To impart flexible, diverse, and learner-centric education that nurtures creativity, critical thinking and lifelong learning for holistic student development.
- To foster a dynamic ecosystem that promotes interdisciplinary research, innovation, and entrepreneurial thinking, contributing to knowledge creation and societal advancement.
- To develop industry-ready graduates through technology-enabled education, experiential learning, international exposure, and innovation-driven skill development that fosters employability and entrepreneurship.
- To nurture globally competent and socially responsible graduates through international collaborations, cross-cultural engagement, and integration of emerging global perspectives.
- To ensure inclusive and equitable education that fosters integrity, ethical values, environmental consciousness, and social commitment for sustainable community development.
- To integrate digital innovation and emerging technologies into teaching, learning, research, and institutional processes for enhanced academic excellence and operational effectiveness.

B.Tech (CSIT) 1st Sem

S No.	Course Category (AICTE)	Course Category (UGC)	BOS	Course Code	Course Name	Type	Academic Learning (AL)			Continuous Internal Examination (CIE)			End Sem Examination (ESE)	Total Marks	Total Credits
							L	T	P	MSE	CA	TOTAL		CIE+ ESE	
1	BS	Major (Core)	ASH	26MA101BSL	Calculus for Engineers	L	3	1	0	80	20	100	100	200	4
2	BS/ ES	Minor Stream	ASH	26PH103BSL/ 26MA201ESL	Engineering Physics/ Discrete Structures & Theory of Logic	L	3	0	0	60	15	75	75	150	3
Blended															
3	ES	Minor Stream	ME/ EEE	26ME101ESB/ 26EE101ESB	Design and Realization/ Introduction to IoT	B	2	0	2	60	15	75	75	150	3
4	ES	Minor Stream	IT	26IT101ESB	Programming For Problem Solving	B	3	0	4	100	25	125	125	250	5
5	ES	Minor Stream	CS	26CS101ESB	Digital Logic Design	B	2	0	2	60	15	75	75	150	3
Lab/Practical															
6	HS	Value Added Courses for all UG	ASH	26AS101HSP	Communication Skills	P	0	0	2	-	25	25	25	50	1
7	PC	Major (Core)	-	-	Industry Integrated Learning Pathways	P	0	0	4	40	10	50	50	100	2
8	BS	Minor Stream	ASH	26PH103BSP	Engineering Physics Lab	P	0	0	2	-	25	25	25	50	1*
9	MC	Value Added Courses for all UG	ASH	26AS102MCP/ 26AS103MCP	Foreign Language/ Indian Knowledge System	P P	0 0	0 0	4 2	-	50	50	-	-	NC
Total Hours : 34 hrs.							**13**	**1**	**20**					**1100**	**22**

*The course may be offered in either the 1st semester or 2nd semester, as per the approved distribution.

Industry Integrated Pathways (IIP)

	Industry Partner: SAP	Industry Partner: AWS	Industry Partner: IBM	Industry Partner: L&T
BOS	**CSE**	**IT**	**IT**	**IT**
1st Sem	Integrated Business Processes in Cloud ERP-I (26CE102PCP)	Web Designing and Cloud Foundations (26IT103PCP)	Web Designing and Cloud Foundations (26IT103PCP)	Web Designing and Cloud Foundations (26IT103PCP)

Foreign Language

Foreign Language	
Course Code	**BOS- PR & IR**
26AS102MCP	Basic Proficiency in Japanese
26AS111MCP	Basic Proficiency in German
26AS112MCP	Basic Proficiency in French
26AS113MCP	Basic Proficiency in Spanish

B. Tech (CSIT) 2nd Sem

S No.	Course Category (AICTE)	Course Category (UGC)	BOS	Course Code	Course Name	Type	Academic Learning (AL)			Continuous Internal Examination (CIE)			End Sem Examination (ESE)	Total Marks	Total Credits
							L	T	P	MSE	CA	TOTAL		CIE+ ESE	
1	ES/ BS	Minor Stream	ASH	26MA201ESL/ 26PH103BSL	Discrete Structures & Theory of Logic/ Engineering Physics	L	3	0	0	60	15	75	75	150	3
Blended															
2	BS	Major (Core)	ASH	26MA102BSB	Linear Algebra for Engineers	B	3	0	2	80	20	100	100	200	4
3	ES	Minor Stream	EEE/ ME	26EE101ESB/ 26ME101ESB	Introduction to IoT/ Design and Realization	B	2	0	2	60	15	75	75	150	3
4	PC	Major (Core)	-	-	Industry Integrated Learning Pathways	B	2	0	2	60	15	75	75	150	3
5	ES	Minor Stream	CSE	26CE201ESB	Data Structures	B	3	0	4	100	25	125	125	250	5
6	ES	Major (Core)	CS	26CS102ESB	Computer Organization and Architecture	B	2	0	2	60	15	75	75	150	3
Lab/Practical															
7	HS	Value Added Courses for all UG	ASH	26AS201HSP	Public Speaking Skills	P	0	0	2	-	25	25	25	50	1
8	BS	Minor Stream	ASH	26PH103BSP	Engineering Physics Lab	P	0	0	2	-	25	25	25	50	1*
9	MC	Value Added Courses for all UG	ASH	26AS103MCP/ 26AS102MCP	Indian Knowledge System/ Foreign Language	P P	0 0	0 0	2 4	-	50	50	-	-	NC
10	PW	Value Added Courses for all UG	CSIT	26CT101PWP	Design Thinking for Innovation and Entrepreneurship	P	0	0	2	-	50	50	-	50	1
Total Hours : 33 hrs.							**15**	**0**	**18**					**1150**	**23**

*The course may be offered in either the 1st semester or 2nd semester, as per the approved distribution.

Industry Integrated Pathways (IIP)

	Industry Partner: SAP	Industry Partner: AWS	Industry Partner: IBM	Industry Partner: L&T
BOS	**CSE**	**IT**	**IT**	**IT**
2nd Sem	Integrated Business Processes in Cloud ERP-II (26CE202PCB)	Python for AI (26IT201PCB)	Python for AI (26IT201PCB)	Python for AI (26IT201PCB)

Foreign Language

Foreign Language	
Course Code	**BOS- PR & IR**
26AS102MCP	Basic Proficiency in Japanese
26AS111MCP	Basic Proficiency in German
26AS112MCP	Basic Proficiency in French
26AS113MCP	Basic Proficiency in Spanish

Theory Courses Detail Syllabus

Course Code: 26MA101BSL	Course Name: Calculus for Engineers	L	T	P	C
		3	1	0	4

Pre-requisite: X+2

Co-requisite: NA

Anti-requisite: NA

Course Objectives:

1. The objective of this course is to familiarize the graduate engineers with techniques of multivariate analysis of real, complex and vector functions in calculus.
2. It aims to impart the knowledge of tools from intermediate to advanced level that will enable them to handle complex problems and its applications so that they would find useful in their disciplines.

Course Outcome: After completion of the course, the student will be able to

1. Apply the concept of partial differentiation in application of homogeneous and composite functions.
2. Apply knowledge of partial differentiation in extrema, series expansion of functions and Jacobians.
3. Apply the concept of vector differentiation in engineering problems.
4. Employ the concept of multiple integration to find the area of bounded region.
5. Apply the concept of analytic and harmonic functions of complex variables in transformation.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	2	-	-	-	1	-	-	-	2
CO2	2	2	2	-	-	-	1	-	-	-	2
CO3	3	2	2	-	-	-	1	-	-	-	1
CO4	2	2	2	-	-	-	1	-	-	-	1
CO5	2	2	2	-	-	-	1	-	-	-	1

Unit 1	Differential Calculus I	09 hours

Introduction of Limits, continuity and differentiability for function of two variables, Partial derivatives, Composite functions, Total derivative, Euler's Theorem for homogeneous functions, Change of variables (except polar form)

Applications: Modeling, analysis, and optimization of multivariable engineering systems; performance evaluation and sensitivity analysis in artificial intelligence, data science, computing systems, communication networks, manufacturing processes, thermal-fluid systems, and other emerging technological domains.

Unit 2	Differential Calculus II	09 hours

Taylor's and Maclaurin's expansion for function of two variables, Jacobians, properties of Jacobian (without proof), Hessian Matrix, Maxima & Minima for a function of several variables, Lagrange's method of multipliers.

Applications: Optimization and analysis of engineering systems including machine learning models, neural networks, resource allocation, electronic circuits, antenna systems, engineering design, structural behavior, and thermal energy systems.

Unit 3	Vector differentiation	09 hours

Scalar point function, Vector point function, Gradient of a scalar field, Directional derivatives, Divergence and curl of a vector function and their application to solenoidal and irrotational vectors respectively

Applications: Modeling and analysis of vector fields and multidimensional systems in artificial intelligence, robotics, computer graphics, electromagnetics, wireless communications, antenna engineering, fluid dynamics, aerodynamics, and autonomous navigation systems.

Unit 4	Multiple Integral	09 hours

Evaluation of double integrals, change of order of integration, Change of variable (double -integral). Application of double integrals to find area of a region.

Applications: Quantitative analysis and computation in image processing, probabilistic AI models, electromagnetic systems, wireless communications, antenna engineering, geometric measurements, heat transfer, and thermal engineering applications.

Unit 5	Complex Variable – Differentiation	09 hours

Functions of complex variable, Limit, Continuity and differentiability, Analytic functions, Cauchy- Riemann equations (Cartesian and Polar form), Harmonic function, Conformal mapping, Mobius transformation.

Application: Coordinate transformations and multidimensional mapping in computer graphics, image processing, secure computing systems, high-frequency communications, electromagnetic wave analysis, fluid dynamics, aerodynamics, and engineering simulations.

	Total Lecture Hours	45 hours

Textbook:

1. V. Ramana, Higher Engineering Mathematics, McGraw-Hill Publishing Company Ltd., 2017
2. B. S. Grewal, Higher Engineering Mathematics, Khanna Publisher, 2020.
3. R K. Jain & S R K. Iyenger, Advanced Engineering Mathematics, Narosa Publishing House 2017.

Reference Books:

1. Dan Hamilton, Calculus 1 - Differentiation and Integration, Hamilton Education Guides 2018.
2. Maurice D. Weir, Joel Hass, Frank R. Giordano, Thomas' Calculus, Pearson, 2002.
3. Peter V. O'Neil, Advanced Engineering Mathematics, Thomson (Cengage) Learning, 2007.
4. E. Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons, 2015.

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
80	20	**100**	**200**
100			

Course Code: 26PH103BSL	**Course Name: Engineering Physics**	**L**	**T**	**P**	**C**
		3	**0**	**0**	**0**

Pre-requisite: Basics Knowledge of Class XII Physics and Mathematics

Co-requisite: NA

Anti-requisite: NA

Course Objectives: Understand the basic principles of quantum physics and their applications in computing. Explain semiconductor physics and the operation of electronic devices. Study the working and applications of optoelectronic devices such as solar cells, LEDs, and photodetectors. Understand the fundamentals and applications of optical fiber communication. Explore advanced materials and their role in modern computing and electronic systems.

Course Outcome: After completion of the course, the student will be able to

1. Apply quantum physics principles to modern computing technologies.
2. Apply semiconductor concepts to determine the characteristics of BJT, FET and MOSFET.
3. Apply optoelectronic principles to solar cells, photodetectors and LEDs.
4. Apply optical-fibre principles in communication systems.
5. Apply material properties to select dielectric, magnetic, smart and piezoelectric materials for computing applications.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	1	2	-	-	-	-	-	-	1	-	2
CO2	2	2	-	-	1	2	-	-	1	-	2
CO3	2	2	1	-	1	2	-	-	1	-	2
CO4	2	2	-	-	2	2	-	-	1	-	2
CO5	2	2	-	-	1	2	-	-	1	-	2

Unit 1 **Quantum Physics** **08 hours**

Inadequacy of classical mechanics, Planck's theory of black body radiation (qualitative), de-Broglie concept of matter waves, Heisenberg's uncertainty principle, Phase velocity and group velocity, Time-dependent and time-independent Schrodinger wave equations, Physical interpretation of wave function, Particle in a one- Dimensional box.

Unit 2 **Semiconductors and Its Devices** **10 hours**

Band Theory of Solids, Fermi-Dirac distribution, Free carrier density (electrons and holes), Conductivity of semiconductors, Fermi level in intrinsic and extrinsic semiconductors, Bipolar junction transistor, p-n-p and n-p-n transistors, Introduction of FET and MOSFET, I-V characteristics.

Unit 3 **Applications of Semiconducting Devices** **10 hours**

Solar Cell: Photovoltaic effect, Construction and working of solar cell, I-V characteristics of solar cell, Applications of solar cells.
Photodetectors: Principle of photodetector and Phototransistor, Construction and working of photodiode and PIN diode, Applications of photodetectors.
Light Emitting Diode (LED): Direct and indirect band gap semiconductors, Differences between homo and hetero junction LEDs, Construction and working of homo junction LED, Characteristics, advantages, and applications of LED.

Unit 4 **Fiber Optics** **09 hours**

Introduction to fiber optics- Total internal reflection, Critical angle of propagation, Acceptance angle, Numerical aperture, V-number (qualitative), Classification of optical fibres: based on number of modes and refractive index profile, Application of optical fiber, Fiber optic sensors.

Unit 5 **Advanced Materials for Computing Systems** **08 hours**

Dielectric materials, Dielectric constant and Polarization of dielectric materials, Magnetic materials (Diamagnetic, Paramagnetic and Ferromagnetic materials), Hysteresis curve and its applications, Smart materials, Piezoelectric materials and its applications.

Total Lecture Hours **45 hours**

Textbook:

1. Arthur Beiser, Concepts of Modern Physics, 7th Edition, McGraw-Hill Education, 2015.
2. Donald A. Neamen, Semiconductor Physics and Devices, 4th Edition, McGraw-Hill Education, 2012.
3. S.O. Kasap, Optoelectronics and Photonics: Principles and Practices, 2nd Edition, Pearson Education, 2012.
4. B.K. Pal, Fundamentals of Fibre Optics in Telecommunication and Sensor Systems, 2nd Edition, New Age International Publishers, 2005.
5. V. Raghavan, Materials Science and Engineering: A First Course, 6th Edition, PHI Learning, 2015.

Reference Books:

1. V.K. Mehta and Rohit Mehta, Principles of Electronics, 12th Edition, S. Chand Publishing, 2020.
2. Ben G. Streetman and Sanjay Banerjee, Solid State Electronic Devices, 7th Edition, Pearson Education, 2015.
3. B.B. Laud, Lasers and Non-Linear Optics, 3rd Edition, New Age International Publishers, 2011.

4. John M. Senior, Optical Fiber Communications: Principles and Practice, 3rd Edition, Pearson Education, 2009.
5. Charles Kittel, Introduction to Solid State Physics, 8th Edition, Wiley India, 2005.

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
60	15	**75**	**150**
75			

Course Code: 26MA201ESL	**Course Name: Discrete Structures & Theory of Logic**	**L**	**T**	**P**	**C**
		3	**0**	**0**	**3**

Pre-requisite: X+2

Co-requisite: NA

Anti-requisite: NA

Course Objectives:

1. The objective of this course is to familiarize the graduate students with the fundamentals of discrete structure and theory of logics.
2. It aims to apply the theory of inferences and graphs in solving the advanced technological problems.

Course Outcome: After completion of the course, the student will be able to

1. Acquire knowledge of sets, relations and functions.
2. Apply fundamental concepts of functions and Boolean algebra in logical reasoning and computational abilities.
3. Employ the rules of propositions, theory of inferences and predicate logic in logical reasoning problems.
4. Understand the concepts of algebraic structures and their applications to apply in critical thinking
5. Apply the concept of graph theory in solving shortest path engineering problems

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	1	1	1	-	-	1	-	-	-	1
CO2	2	1	1	-	-	-	1	-	-	-	1
CO3	2	1	1	1	-	-	1	-	-	-	1
CO4	2	2	1	-	-	-	1	-	-	-	1
CO5	2	2	2	-	-	-	1	-	-	-	2

Unit 1 **Sets, Relations & Functions** **09 hours**

Set Theory& Relations: Introduction, Operations on sets. Relations: definition, Operations on relations, properties of relations, Composite relations. **Functions**: Definition, classification of functions, Operations on functions.
Applications: Mathematical foundations for data organization, relationship modeling, functional mappings, computational reasoning, and problem-solving in modern computing and intelligent systems.

Unit 2 **Lattices & Boolean Algebra** **09 hours**

POSET & Lattices: POSET, Hasse Diagram, Lattice: definition & properties – Bounded, Complemented, Distributed, Modular and Complete lattice. **Boolean Algebra:** Introduction, Axioms and theorems of Boolean algebra, properties, simplification of Boolean expressions, CNF and DNF, Karnaugh maps (upto three variables).
Application: Modeling and analysis of ordered structures, logical systems, digital circuits, computational workflows, and intelligent decision-support systems.

Unit 3 **Theory of Logic** **09 hours**

Theory of Logics: Proposition, Truth tables, Tautology, Contradiction, Algebra of proposition, Theory of Inference. Predicate Logic: First order predicate, quantifiers, Inference theory of predicate logic.
Application: Foundations for logical reasoning, decision support, automated inference, knowledge-based systems, and intelligent computing applications.

Unit 4 **Algebraic Structures** **09 hours**

Algebraic Structures: Groups, subgroup , order of a group, Abelian and Cyclic groups with properties, cosets, Lagrange's theorem (without proof), Permutation group.
Application: Modeling of symmetries, transformations, permutations, and algebraic operations in computing, communication, and information systems.

Unit 5 **Graph Theory** **09 hours**

Graphs: Definition and terminology, matrix representation of graphs, handshaking theorem, types of graphs, bipartite graph, Isomorphism of graphs, Recurrence relation (homogeneous case only).
Application: Representation and analysis of networks, relationships, and recursive structures in computational and engineering systems.

Total Lecture Hours **45 hours**

Textbook:

1. Trembley, J.P & R. Manohar, "Discrete Mathematical Structure with Application to Computer Science", Tata McGraw Hill. 1997 2nd edition Reprint 2017
2. Swapan Kumar Sarkar, A Textbook of Discrete Mathematics, S Chand Publishing.

Reference Books:

1. C. L. Liu, Elements of Discrete Mathematics: A Computer Oriented Approach, McGraw Hill. 4th edition (Paperback 2017)
2. Narsingh Deo, Graph Theory with Applications to Engineering and Computer Science, Phi Learning
3. E.R. Scheinerman, Mathematics: A Discrete Introduction, Brooks/Cole, 3rd edition
4. Thomas Koshy, Discrete Mathematics with Application, Elsevier Pub. 2004
5. Kenneth H. Rosen, Discrete Mathematics and Its Applications, McGraw-Hill
6. Liptschutz, Seymour, "Discrete Mathematics", McGraw Hill, 3rd edition (Paperback 2017)
7. B. Kolman, R.C. Busby, and S.C. Ross, Discrete Mathematical Structures, Prentice Hall, 3rd edition.

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
60	15	**75**	**150**
75			

Blended Courses Detail Syllabus

Course Code: 26ME101ESB	Course Name: Design and Realization	L	T	P	C
		2	0	2	3

Pre-requisite: Basic computer knowledge

Co-requisite: NA

Anti-requisite: NA

Course Objectives:

1. Develop skills in computer-aided engineering graphics for 2D and 3D modelling.
2. Improve visualization and design interpretation abilities.
3. Understand fundamentals of CNC machining and 3D printing processes.
4. Enable realization of designs using CAD, CNC machining, and 3D printing.

Course Outcome: After completion of the course, the student will be able to

1. Create 2D drawings and 3D models using CAD tools.
2. Interpret and develop engineering designs from given requirements.
3. Explain basics of 3D printing processes.
4. Explain basics of CNC machining.
5. Produce components using CAD models through CNC machining and 3D printing.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	2	-	-	-	2	-	3
CO2	2	-	2	-	3	-	-	-	2	-	3
CO3	2	-	2	-	3	-	-	2	2	-	3
CO4	2	-	2	-	3	-	-	2	2	-	3
CO5	-	-	-	-	2	-	-	2	2	-	3

Unit 1	Computer-aided Design- 2D Drawing	20 hours

Introduction to Engineering drawing, First and third angle projection. Orthographic Projection: projection of point, Projection of solids, Principles of isometric projection, Introduction to computer-aided design, Structure of CAD Software's, Understanding the GUI and tools of Autodesk Fusion 360, 2D Drawing/Sketching and Editing Commands, Understanding geometric constraints.

Hands-on: Problem Solving on 2D sketching on Autodesk Fusion 360.

Unit 2	Computer-aided Design- 3D Modelling	20 hours

Introduction to 3D modelling, understand 3D commands in Autodesk Fusion, including extrude, revolve, loft, and sweep etc., creating 3D parts from 2D sketches, Part editing and modification, Assembly modelling and constraints, Generating exploded views and animations.

Hands-on: Problem Solving on 3D sketching on Autodesk Fusion 360.

Unit 3	3D Printing	07 hours

Introduction to 3D printing, 3D scanning, Preparing 3D models for printing, AM file formats, slicing-uniform slicing and adaptive slicing, Process-path generation, Understanding the interface of AM Software: Ultimaker Cura or similar.

Hands-on: Designing a 3D model and print using 3D printer.

Unit 4	Additive Manufacturing	07 hours

Design considerations for 3D printing (supports, overhangs, infill), Post processing of 3D printed parts, Introduction to 3D printing Techniques - Liquid based, Solid state and Powder based techniques for 3D printing.

Hands-on: Designing a 3D model and print using 3D printer.

Unit 5	Computer Aided Manufacturing	06 hours

Introduction to CNC Machining, Types of CNC machines, Components of a CNC machine (e.g., controller, spindle, axes, ATC), CNC machine configurations (e.g.,3-axis,5-axis). CNC Programming Basics: Introduction to G-code and M-code programming, 3D modelling for CNC machining, understanding file formats for CNC machines.

Hands-on: Designing a 3D model and fabrication using a CNC machine.

Total Lecture Hours	60 hours

Textbook:

1. Engineering Graphics & Design, P. S. Gill.
2. Computer-Aided Graphics and Design, Daniel L. Ryan
3. Computer-Aided Design and Manufacturing by M. Groover.

Reference Books:

1. Engineering Graphics With AUTOCAD, Kulkarni D.M
2. An Introduction to 3D Printing by Victoria Zukas and Jonas A. Zukas
3. Computer Aided Manufacturing, P.N. Rao, N.K. Tewari, T.K. Kundra

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
60	15	75	150
75			

Course Code: 26EE101ESB	Course Name: Introduction to IoT	L	T	P	C
		2	0	2	3

Pre-requisite: Basic computer knowledge

Co-requisite: NA

Anti-requisite: NA

Course Objectives:

1. The course aims to provide exposure to the applications of IoT in smart cities and industrial applications.
2. It aims to train the students in the basic concepts of the Embedded C and Controllers.
3. It aims students to IoT communication, cloud platforms, Blynk, ThingSpeak, HTTP, and MQTT.
4. It aims students to design, develop, and demonstrate smart IoT-based monitoring and automation applications.

Course Outcome: After completion of the course, the student will be able to

1. Understand the basic concepts of sensors and transducers.
2. Understand basics of embedded system and different IoT boards.
3. Apply basic operations and programming techniques of IoT devices.
4. Understand IoT communication, cloud platforms.
5. Design and demonstrate smart IoT-based applications or mini-projects.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	-	-	-	2	2	-	-	-	-	2
CO2	2	-	2	-	2	2	-	2	-	-	2
CO3	3	-	3	2	3	2	-	2	-	-	2
CO4	3	2	3	3	3	2	-	2	-	-	2
CO5	3	2	3	3	3	2	-	2	-	-	2

Unit 1 **Sensing Devices & Transducers** **12 hours**

Sensors & Transducer: Definition, Types & selection criterion of sensors, Classification of Sensors & Transducer based on principle of operation, Fundamentals & Applications of Potentiometer, Fundamentals & Applications of strain gauge.

Hands-on/Case Study/ Mini-Project/ Problem solving:

- Understanding Sensors and Transducers: Types, Operation Principles, and Selection Criteria.
- Role of Sensors and Transducers in Internet of Things (IoT) Systems.
- Hands-on Introduction to commonly used real world IoT Sensors.
- Hands-on Introduction to different controllers used in IoT.

Unit 2 **Embedded Systems Fundamentals** **12 hours**

Introduction to Embedded C: Interfacing Basics, Digital I/O, Analog I/O, Introduction to Arduino (ATmega328P), Arduino board components and architecture, Interfacing DHT11 with Arduino.

Hands-on/Case Study/ Mini-Project/ Problem solving:

- Understanding the Architecture and Pin Configuration of Arduino Boards.
- Analyse Digital signal data acquisition using Arduino.
- Explore Digital signal generation using Arduino.
- Analyse Analog signal data acquisition using Arduino.
- Explore Analog signal generation using Arduino.

Unit 3 **IoT Board** **12 hours**

Introduction to IoT in Modern Industry Applications, Basic Operations of IoT, Basics of ESP 8266 programming, Introduction to Blynk IoT, Interfacing with Different types of Sensors: Touch Sensor, Alcohol Sensor (MQ 3), LPG Sensor (MQ 6), Relay, Light Dependent Resistor (LDR), IR (Infrared) Sensors and PIR (Passive Infrared) Sensors.

Hands-on/Case Study/ Mini-Project/ Problem solving:

- Understanding the Architecture and Pin Configuration of ESP8266.
- Analyse Digital / Analog signal data acquisition using ESP8266.
- Explore Digital / Analog signal generation using ESP8266.
- Real-Time Data Logging Using ESP8266 and Arduino.

Unit 4 **IoT Communication and Cloud Platforms** **12 hours**

IoT architecture, device-to-cloud communication, Wi-Fi-based IoT using ESP8266, basics of HTTP and MQTT, introduction to Blynk and ThingSpeak platforms, real-time data monitoring, cloud dashboard creation, and remote control of IoT devices.

Hands-on/Case:

- Study temperature and humidity monitoring using ESP8266.
- Relay-based load control using Blynk, cloud data logging using ThingSpeak, and smart home automation case study.

Unit 5 **Smart Sensor-Based IoT Applications** **12 hours**

Intelligent Sensors: General Structure of smart sensors & its components, Case study of Air Quality Monitoring System, Case study of Soil Health Monitoring System, Case study of Water Quality Monitoring System.

Hands-on/Case Study/ Mini-Project/ Problem solving:

- Designing a Lighting Control System using LDR.
- Designing a Multi-Sensor Alert System Using Touch, IR, PIR and Arduino.
- Object Detection Using Ultrasonic Sensors with Arduino and ESP.
- Building IoT Applications with Blynk: Monitoring Temperature and Humidity with DHT11 Sensor.
- Building IoT Applications with Blynk: Smart Home Automation Using ESP8266 and Blynk.
- Building a Soil Health Monitoring system using NPK sensor.

Total Lecture Hours	60 hours

Textbook:

1. Raj Kamal, "Internet of Things: Architecture and Design Principles", McGraw Hill Education (India) Private Limited CHENNAI.
2. Waldemar Nawrocki, "Measurement Systems and Sensors", Artech House Boston, London.
3. Rajkumar Buyya and Amir Vahid Dastjerdi, "Internet of Things: Key Applications and Protocols" Elsevier.

Reference Books:

1. Pethuru Raj and Anupama C. Raman. "The Internet of Things: Enabling technologies, platforms, and use cases". Auerbach Publications.
2. "The Internet of Things: Do-It-Yourself at Home Projects for Arduino, Raspberry Pi and Beagle Bone Black" by Donald Norris, McGraw-Hill Education TAB.

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
60	15	**75**	**150**
75			

Course Code: 26IT101ESB	Course Name: Programming For Problem Solving	L	T	P	C
		3	0	4	5

Pre-requisite: Basic knowledge of computers, algorithms, flowcharts, and programming concepts.

Co-requisite: NA

Anti-requisite: NA

Course Objectives:

1. To introduce the fundamental concepts of programming using C++.
2. To develop problem-solving ability using algorithms, pseudo-code, decision-making, and iterative constructs.
3. To enable students to apply arrays, pointers, functions, strings, structures, and STL components for solving computational problems.
4. To develop modular programming skills using functions, parameter passing, and recursion.
5. To prepare students to write readable, structured, and efficient programs for basic engineering and data-processing applications.

Course Outcome: After completion of the course, the student will be able to

1. Apply basic C++ programming constructs such as input-output, operators, decision-making, and looping statements to solve computational problems.
2. Implement C++ programs using arrays, pointers, and dynamic memory allocation for solving data-processing problems.
3. Develop modular programs using functions, parameter passing techniques, storage classes, and recursion.
4. Design and implement C++ programs using strings and structures for record-based and text-processing applications.
5. Apply Standard Template Library components such as vector, array, iterators, and STL algorithms to solve programming problems efficiently.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2	-	1	-	-	-	1	-	2
CO2	3	3	2	1	2	-	-	-	1	-	2
CO3	3	3	3	1	1	-	-	-	1	-	2
CO4	3	2	3	1	2	-	-	-	1	-	2
CO5	3	2	2	-	2	-	-	-	1	-	2

Unit 1	Introduction to Programming and Control Structures (Using C++)	21 hours

Programming Fundamentals: Structure of a C++ program, writing and executing the first program, compilation and execution process. **Tokens in C++:** Keywords, identifiers, variables, constants, data types, character set. **Operators and Expressions:** Arithmetic, relational, logical, assignment, unary, conditional, bitwise, and comma operators; operator precedence and associativity; type conversion and type casting. **Decision-Making Statements**: if, if-else, nested if, if-else-if ladder, switch-case. **Iterative Statements**: while, do-while, for loop, nested loops. **Control Transfer Statements**: break, continue.

Programming Practices: Code readability, naming conventions, indentation, and documentation using comments. Problem solving using decision-making and iterative statements, including pattern-based problems.

Suggested Problem for LAB: Programming problems shall be based on the concepts covered in this unit, including C++ basics, input-output, operators, decision-making, loops, nested loops, control transfer statements, and logic-building problems.

Indicative Application-Based Problems: ATM transaction and security simulation, student result processing and eligibility system, electricity billing and tariff management system, number theory and mathematical utility system, pattern/series logic-building toolkit, and basic

login/access control system using decision-making and iterative constructs.

Unit 2	Arrays & Pointers	21 hours

Arrays: One-dimensional arrays: declaration, initialization, traversal, insertion, and deletion. **Multi-dimensional Arrays:** Two-dimensional arrays and basic operations. **Pointers:** Introduction, declaration, pointer arithmetic, pointers and arrays, pointer to pointer, arrays of pointers. **Dynamic Memory Allocation:** using new and delete.

Suggested Problem for LAB: Array operations, searching and sorting techniques, matrix operations, pointer-based processing, pointer arithmetic, and dynamic memory allocation using new and delete.

Indicative Application-Based Problems:

Student marks and rank analysis system, inventory stock processing system, matrix-based computational toolkit, pointer-based array processing utility, and dynamic list management system.

Unit 3	Functions & Recursion	21 hours

Functions: Need for modular programming, function declaration, definition, prototype, and function calling in C++. **Parameter Passing Techniques:** Call by value and call by reference (using reference variables). Passing arrays to functions. **Scope and Lifetime of Variables:** Block scope, function scope, and introduction to global scope. **Storage Classes:** auto, register, extern, static. **Recursion:** Concept, working mechanism, and simple applications.

Suggested Problem for LAB:

Function-based programs such as calculator and student result processing, parameter passing, array processing using functions, storage class demonstrations, and recursive problems such as factorial, Fibonacci, GCD, and power calculation.

Indicative Application-Based Problems:

Modular calculator application, student result processing using functions, array processing toolkit, number theory utility using functions, recursive mathematical computation system, and function-based searching and sorting applications.

Unit 4	Strings and Structures	21 hours

Strings in C++: Use of string data type, declaration and initialization of strings, input and output operations on strings. **Basic string operations**: concatenation, length, comparison, and substring. **Structures in C++:** Structure declaration, definition, initialization, accessing members, nested structures, arrays of structures, and passing structures to functions.

Suggested Problem for LAB:

String processing problems, text analysis, student/employee record management using structures, arrays of structures, nested structures, and passing structures to functions.

Indicative Application-Based Problems:

Student record management system, employee payroll system, library book record system, text analysis utility, contact management system, and product/inventory record management system using strings and structures.

Unit 5	Standard Template Library (STL)	21 hours

Introduction to STL: Concept and importance of STL. Overview of STL components: containers, iterators, and algorithms. **Containers:** Usage of commonly used STL containers such as vector and array. Declaration, initialization, traversal, insertion, deletion, element access, and size-related operations. **Iterators:** Concept of iterators and their role in container traversal. Basic traversal using begin (), end (), rbegin (), and rend (). **Algorithms:** Introductory use of STL algorithms such as sort(), find(), reverse(), count(), min_element(), and max_element() for searching, sorting, reversing, counting, and finding minimum/maximum elements.

Suggested Problem for LAB:

STL-based programs using vector, array, iterators, sort(), find(), and simple operations such as traversal, searching, sorting, reversing, duplicate removal, and frequency counting.

Indicative Application-Based Problems:

Student marks processing using vector, name sorting and searching application, inventory list management, duplicate data removal utility, frequency counting system, fixed-size data processing using array, and basic record processing using STL containers and algorithms.

Total Lecture Hours	105 hours

Textbook:

1. E. Balagurusamy, Object-Oriented Programming with C++, 8th Edition, McGraw Hill Education.
2. Bjarne Stroustrup, Programming: Principles and Practice Using C++, 3rd Edition, Addison-Wesley.
3. Herbert Schildt, C++: The Complete Reference, 4th Edition, McGraw Hill Education.

Reference Books:

1. Stanley B. Lippman, Josée Lajoie, and Barbara E. Moo, C++ Primer, 5th Edition, Addison-Wesley.
2. Robert Lafore, Object-Oriented Programming in C++, 4th Edition, Pearson Education.
3. Scott Meyers, Effective C++, 3rd Edition, Addison-Wesley.

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
100	25	**125**	**250**
125			

Course Code: 26CS101ESB	Course Name: Digital Logic Design	L	T	P	C
		2	0	2	3

Pre-requisite: Basic knowledge of engineering concepts and logical problem-solving.

Co-requisite: NA

Anti-requisite: NA

Course Objectives:

1. Provide a foundation in digital logic concepts and techniques used in the design of digital systems.
2. Develop the ability to analyse and design combinational and sequential circuits.
3. Familiarize students with performance aspects such as timing, delays, and hazards in digital systems.
4. Introduce modern design practices using HDL tools for modelling and implementation of digital circuits.

Course Outcome: After completion of the course, the student will be able to

1. Apply Boolean algebra and minimization techniques to design and simplify digital logic circuits.
2. Construct combinational logic circuits for arithmetic, coding, and data selection applications.
3. Construct synchronous sequential circuits using flip-flops and state machine concepts.
4. Demonstrate the operation of registers, counters, and timing characteristics in digital systems.
5. Implement digital system components using Datapath elements, programmable logic devices, and HDL.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	-	-	-	-	-	-	-	-
CO2	3	2	3	-	-	-	-	-	-	-	-
CO3	3	2	3	-	-	-	-	-	-	-	-
CO4	2	3	1	1	-	-	-	-	-	-	-
CO5	2	1	2	1	3	1	-	-	-	-	2

Unit 1 | Fundamentals of Digital Logic | 12 hours

Number systems (binary, octal, hex), Binary arithmetic & codes (BCD, Gray), Code Converters, Boolean algebra & theorems, SOP, POS forms, Minimization techniques: K-Maps (including don't care conditions), Quine McCluskey method, Logic Gates: AND, OR, NOT, NAND, NOR, XOR, Universal gates implementation

Indicative Programs:

- Verification of basic logic gates (AND, OR, NOT, NAND, NOR, XOR) using truth tables.
- Implementation of basic gates using universal gates (NAND and NOR).
- Simplification of Boolean expressions using K-Maps (up to 4 variables).
- Realization of Boolean expressions in SOP and POS forms.
- Design and implementation of code converters (Binary ↔ Gray, BCD).

Unit 2 | Combinational Logic Circuits | 12 hours

Adders, subtractors, Binary adder/subtractor, Design of 4-bit carry look ahead adders and delay analysis, Multiplexers, Demultiplexers, Encoders, Decoders.

Sample Programs:

- Design and implementation of Half Adder and Full Adder.
- Design and implementation of Half Subtractor and Full Subtractor.
- Design of a 4-bit Ripple Carry Adder.
- Design of a 4-bit Carry Lookahead Adder.
- Implementation of 4:1 Multiplexer and realization of Boolean functions using MUX.
- Design and implementation of Demultiplexer.
- Design of Encoder and Priority Encoder.
- Design of Decoder (e.g. 3-to-8) and its applications.
- Design of a 4-bit Magnitude Comparator.

Unit 3 | Sequential Logic Circuits and Design | 12 hours

SR Latch (NAND/NOR based), Latch vs Flip-Flop, SR, JK, D and T flip flops, flip-flop excitation tables, level triggering and edge triggering, state diagram, state table, state reduction, state assignment, Synchronous sequential circuit design.

Sample Programs:

- Implementation of SR Latch using NAND/NOR gates.
- Design and verification of SR, JK, D, and T Flip-Flops.
- Conversion between Flip-Flops (e.g., JK to D, D to T).
- Design and implementation of a simple synchronous sequential circuit using state diagram and state table.
- Design of a synchronous sequential circuit using state diagram and state table.

Unit 4 | Registers, Counters and Timing Analysis | 12 hours

Registers, Shift registers (SISO, SIPO, PISO, PIPO), Asynchronous and Synchronous counters, Modulo-N counter, Timing analysis (propagation delay, setup time, hold time)

Sample Programs:

- Design and implementation of Shift Registers (SISO, SIPO, PISO, PIPO).
- Analysis of propagation delay in digital circuits (simulation-based).
- Design of Asynchronous (Ripple) Counter.
- Design of Synchronous Counter.
- Design of Modulo-N Counter.

Unit 5	**Digital System Design and HDL**	**12 hours**

Datapath Components: ALU, Programmable Logic: PLA, PAL, CPU Bus Implementation: MUX, Tri-state Buffers, Introduction to HDL: Verilog/VHDL

Sample Programs:

- Design of a simple 4-bit ALU.
- Implementation of basic logic circuits using HDL (Verilog/VHDL).
- Design of combinational circuits (e.g., Adder/Decoder) using HDL.
- Design of sequential circuits (e.g., Counter/Flip-Flop) using HDL.

	Total Lecture Hours	**60 hours**

Textbook:

1. Digital Design, M. Morris Mano and Michael D. Ciletti, 5th Edition, Pearson Education.
2. Fundamentals of Digital Circuits, A. Anand Kumar, 3rd Edition, PHI Learning.
3. Digital Electronics: Principles, Devices and Applications, Anil K. Maini, Wiley India.

Reference Books:

1. Fundamentals of Logic Design, Charles H. Roth Jr. and Larry L. Kinney, 7th Edition, Cengage Learning.
2. Digital Design and Computer Architecture, David Money Harris and Sarah L. Harris, 2nd Edition, Morgan Kaufmann.
3. Modern Digital Electronics, R. P. Jain, 4th Edition, McGraw-Hill Education.

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
60	15	**75**	**150**
75			

Course Code: 26MA102BSB	**Course Name: Linear Algebra**	**L**	**T**	**P**	**C**
		3	**0**	**2**	**4**

Pre-requisite: X+2

Co-requisite: MATLAB

Anti-requisite: NA

Course Objectives:

1. The objective of this course is to develop a strong foundation in linear algebra and to impart the knowledge of tools from intermediate to advanced level of mathematics.
2. Students will be equipped with the necessary skills to apply linear algebra to solve complex engineering problems.

Course Outcome: After completion of the course, the student will be able to

1. Understand the concept of vector space and subspaces.
2. Apply elementary transformation to solve system of Linear equations.
3. Understand the concept of complex matrix and Eigen vectors
4. Apply the concept of matrix factorization in data decomposition.
5. Explore the concept of linear transformations and inner product in engineering applications.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**
CO1	3	3	3	-	-	-	1	-	-	-	2
CO2	2	2	2	-	-	-	1	-	-	-	1
CO3	2	2	2	-	-	-	1	-	-	-	1
CO4	2	2	2	-	-	-	1	-	-	-	1
CO5	2	2	2	-	-	-	1	-	-	-	1

Unit 1	**Vector Spaces**	**15 hours**

Definition of Field, Introduction to Vector Spaces, Basic Properties of Vector Spaces, Sub spaces, Linear dependence and independence of vectors, Basis and Dimension.

Application: Representation, transformation, and analysis of high-dimensional data in modern computing, artificial intelligence, and data-driven systems.

Hands on activities:

- Understand the principles of Field, vector space, Basis and Dimension
- Represent text documents or images as vectors using MATLAB/Python.
- Visualize high-dimensional data in reduced vector spaces.

Unit 2	Matrix Algebra-I	15 hours

Introduction to Matrices, Elementary Transformation, Inverse by E- transformation, Rank of a Matrix by Echelon Form, Solution of system of linear Equations by direct method (Gauss Elimination Method), and Iterative method (Gauss Seidel Method).

Application: Matrix-based modeling and solution techniques for data analysis, optimization, computer graphics, intelligent systems, and computational engineering applications.

Hands on activities:

- Obtain the inverse of a square matrix using MATLAB.
- Obtain the Rank of a matrix using MATLAB
- Write a MATLAB program to obtain the solution of system of Linear Equation

Unit 3	Matrix Algebra-II	15 hours

Hermitian, Skew Hermitian, Unitary Matrices. Eigen Values & Eigen Vectors, Cayley Hamilton theorem (without proof) and its applications.

Application: Analysis and transformation of high-dimensional data in machine learning, computer vision, signal processing, and computational intelligence.

Hands on activities:

- Evaluate Eigen Values and Eigen Vectors of a square matrix using MATLAB.

Unit 4	Applied Matrix Algebra	15 hours

Similarity transformation, Diagonalization of matrix, Eigen Value Decomposition and Singular Value Decomposition, Matrix factorization, LU Decomposition (Dolittle method).

Applications: Advanced matrix-based methods for data analysis, optimization, intelligent computing, and large-scale computational systems.

Hands on activities:

- Write a MATLAB program to diagonalize the square matrix.
- Decompose the square matrix using Eigen Value decomposition method in MATLAB.
- Factorize the m×n matrix by singular value decomposition method in MATLAB.
- Decompose the square matrix in lower and upper triangular matrices by Dolittle's method in MATLAB.

Unit 5	Linear Transformation & Inner Product Space	15 hours

Linear transformation, Matrix representation of Linear Transformation, Kernel and Range, Rank Nullity Theorem (without proof). Introduction to inner product space, Norm of a vector, Orthogonality and orthonormality of vectors, Gram Schmidt orthogonalization process.

Applications: Transformation, representation, and analysis of multidimensional data in computing, intelligent systems, graphics, and communication technologies.

Hinds on activities:

- Write a MATLAB code to represent Linear Transformation in matrix form for standard basis vectors.
- Normalize the vector using MATLAB.

	Total Lecture Hours	75 hours

Textbook:

1. Hoffman, K. and Kunze, R., "Linear Algebra", Pearson Education (Asia) Pvt. Ltd/ Prentice Hall of India, 2015
2. Nair, M.T. & Singh A., Linear Algebra, Springer, 2019.
3. Strang, Gilbert, Linear Algebra and its Application, Cengage Learning, 4th edition, 2005.
4. Jain, M.K., Iyengar, S.R.K. and Jain R.K., Numerical Methods, New Age International Publishers, 2019.

Reference Books:

1. Schaum's Outline of Linear Algebra, McGraw Hill Education 2017
2. Strang, G., "Linear Algebra and Its Applications", Thomson Learning Asia Pvt. Ltd.4th edition ,2005.
3. Lay, Dand C., "Linear Algebra and Its Applications" Pearson Education Limited, 6th edition 2020.
4. Richard, L. Burden, J. Douglas Faires, and Annette Burden, Numerical Analysis, Cengage Learning, 10th edition, 2015.
5. Sastry S. S. "Introductory Methods of Numerical Analysis", PHI, 3rd edition 2002.

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
80	20	**100**	**200**
100			

Course Code: 26CE201ESB	Course Name: Data Structures	L	T	P	C
		3	0	4	5

Pre-requisite: The course requires a background in mathematics and sufficient programming skills.

Co-requisite: NA

Anti-requisite: NA

Course Objectives:

1. To provide a deep understanding of fundamental data structures and their applications.
2. To provide expertise in the efficient implementation of physical and logical data structures.
3. To provide insight into the working of searching and sorting algorithms.
4. To develop the analytical ability for solving real-world problems using data structures.

Course Outcome: After completion of the course, the student will be able to

1. Use the concept of the array in searching and sorting algorithms.
2. Illustrate the concept of Dynamic Memory Allocation for operations on linked lists.
3. Analyze different recursion techniques using stack.
4. Analyze the fundamental concept of queues.
5. Apply the knowledge of tree and binary tree structures for problem solving.

CO-PO Mapping(Scale1: Low, 2:Medium, 3:High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	1	1	-	1	1	-	-	-	-	2
CO2	3	2	2	1	1	1	-	-	-	-	2
CO3	3	2	2	1	1	1	-	-	-		2
CO4	3	2	2	1	1	1	-	-	-	-	2
CO5	3	2	1	-	1	1	-	-	-	-	2

Unit 1 | Introduction | 15 hours

Basic Terminology, Types and application of Data Structures, Algorithm, Efficiency of an algorithm, Time space trade off and complexity, asymptotic notation. Array: Single and Multidimensional Arrays, Representation of Arrays: Row Major Order, and Column Major Order, Derivation of Index Formulae for 1-D,2-D,3-D and n-D Array Application of arrays, Sparse Matrices, and their representations, arithmetic operations on matrices. Searching: Linear search, Binary Search, Indexed Sequential Search, Sorting: Insertion Sort, Bubble sort, Selection sort Application Area: Matrix, Dynamic Programming, Radix Sort, Bucket Sort Buffer or Cache, Stack & Queue, Graph Representation, Tree, Image Processing, Signal Processing, Databases, Web Search Engines, Networking Routing.

Problem solving:

1. Write a program to find the sum of elements of positive and negative elements of a one- dimensional array.
2. Given an integer array nums, write a program to print true if any value appears at least twice in the array, and return false if every element is distinct.
3. Nirobi has given a matrix C of size N x M to Rio. Also, Rio is given the position of submatrix as X1, Y1, and X2, Y2 inside the matrix. Now Rio needs to find the sum of all elements inside that submatrix. Can you help Rio in completing the task assigned by Nirobi?
4. Write a program to implement linear search on an array of integers. Return the index of the target if found, else return -1.
5. Given a sorted array of distinct integers and a target value, write a program to print the index if the target is found. If not, return the index where it would be if it were inserted in order.
6. Write a program to convert a 2D matrix into a triplet representation if it is sparse, and reconstruct the original matrix from the triplet form.
7. Write a program to sort an array using bubble sort. Display the array after each pass and count the number of swaps and comparisons.
8. Write a program to sort an array using selection sort. Display the intermediate steps and total comparisons made.
9. Write a program to sort the given elements using insertion sort technique.

Unit 2 | Linked List | 15 hours

Singly Linked Lists, Doubly Linked List, Circularly Linked List, Operations on a Linked List. Insertion, Deletion, Traversal, Reversing, Polynomial Representation and Addition. Generalized Liked list.

Application Area: Symbol table implementation, Memory Management, Tries, Tree, Graph, Music and Video Playlists, Undo/Redo Functionality, Hash Tables and Collision Resolution

Problem solving:

1. You are given the pointer to the head node of a linked list and an integer to add to the list. Create a new node with the given integer. Insert this node at the tail of the linked list and return the head node of the linked list formed after inserting this new node. The given head pointer may be null, meaning that the initial list is empty.
2. Given the head of a linked list, write a program to determine if the linked list has a cycle in it or not.
3. Write a function to reverse a singly linked list. Given the head of the linked list, modify the list in-place to reverse its order and return the new head of the reversed list.
4. Write a function to insert a given value into the middle of a singly linked list. If the list has an even number of nodes, insert the value after the first half. Return the head of the modified linked list.
5. Write a function to delete every alternate node from a given doubly linked list, starting with the second node. Modify the list in place and return the head of the updated list.
6. Write a program to create a circular singly linked list, insert elements, and traverse the list in a circular manner starting from the head node.
7. Write a program to perform deletion operations (beginning, end, specific position) in a circular singly linked list and display the result.
8. Write a function to reverse a singly linked list in groups of size k. Given the head of the linked list and an integer k, reverse the nodes of the list k at a time. If the number of nodes in the last group is less than k, leave the remaining nodes as they are.
9. Write a program to represent a polynomial using a singly linked list, where each node contains a coefficient and exponent.
10. Write a program to add two polynomials represented using singly linked lists. Output the resulting polynomial after addition.

Unit 3 | Stack | 15 hours

Abstract Data Type, Primitive Stack operations: Push & Pop, Array and Linked List Implementation of Stack, Application of stack: Prefix and Postfix Expressions, Evaluation of postfix expression, Tail recursion, Head Recursion, Nested recursion, Removal of recursion. Problem solving using iteration and recursion with examples such as Fibonacci numbers, and Hanoi towers. Trade-offs between iteration and recursion.

Application Area: Function Call Stack, Optimal Parentheses Problem in Matrix multiplications, Backtracking, Depth-First Search, Parsing and Compiler Design, Process Control Block, Memory Management.

Problem solving:

1. Write a program to implement a stack using an array with basic operations: push, pop and display.

2. Write a program to implement a stack using a linked list. Include stack operations: PUSH, POP, and DISPLAY.
3. Write a program to convert a valid infix expression to postfix notation using a stack.
4. Write a program to convert a valid infix expression to prefix notation using a stack.
5. Write a program to evaluate a postfix expression using a stack. Support basic arithmetic operators (+, -, *, /).
6. Write a recursive program that demonstrates tail recursion with a simple function like factorial calculation.
7. Write a recursive program that demonstrates head recursion to print numbers from n down to 1.
8. Write a program to generate Fibonacci numbers using both iteration and recursion. Compare and comment on performance and memory usage.
9. Write a recursive program to solve the Tower of Hanoi problem for n disks and print the sequence of moves.
10. Simulate the computation of factorial using a stack (non-recursive approach). Show the stack at each stage of the computation.

Unit 4	**Queue**	**15 hours**

Operations on Queue: Create, Add, Delete, Full and Empty, Circular queues, Array and linked implementation of queues, Double Ended queue.

Application Area: Job Scheduling, Breadth-First Search, Search Trees and Binary Search Tree, Database Operations, Customer Service, Web Server Request Handling, Buffering and Data Streaming, Traffic Management.

Problem solving:

1. Write a program to implement a linear queue using an array with operations: ENQUEUE, DEQUEUE, DISPLAY, and check for OVERFLOW and UNDERFLOW.
2. Write a program to implement a queue using a linked list. Perform ENQUEUE, DEQUEUE, and DISPLAY operations dynamically.
3. Write a program to implement a circular queue using an array and support the basic operations: insert, delete, and display queue elements with proper wrap-around logic.
4. Write a program to implement a circular queue using linked list, handling both front and rear pointers correctly.
5. Write a program to implement a double-ended queue (de-queue) using an array or linked list. Support insertions and deletions from both ends.
6. Use a dequeue to check whether a given string is a palindrome. You can insert and remove characters from both ends.

Requirements:

- Insert characters of string into dequeue
- Remove one character from front and rear in each iteration
- If all characters' match, it's a palindrome

Sample Input: "radar"

Expected Output: "The string is a palindrome"

Unit 5	**Trees**	**15 hours**

Binary Tree and Its array and linked list representation, Strict Binary Tree, Complete Binary Tree, Tree Traversal algorithms: In- order, Pre-order and post-order, level order, Constructing Binary Tree from given Tree Traversal, BST Operation: Searching, Insertion, Deletion, Threaded Binary Trees, Traversals in Threaded Binary Trees, Heaps, Heap Sort.

Application Area: Dictionary Implementation, Compiler Design, Graph Algorithms, In order Traversal Optimization

Problem solving:

1. Write a program to create a binary tree using linked representation. Implement insertion of nodes manually and display the structure using level-order traversal.
2. Write a program to implement a binary tree using array representation. Provide functions to insert and retrieve the left and right child of a given node.
3. Given a binary tree, write a program to check whether it is a strict binary tree (every node has 0 or 2 children) and whether it is a complete binary tree.
4. Write a program to implement recursive and non-recursive versions of in-order, pre-order, and post-order traversal of a binary tree.
5. Write a program to construct a binary tree from the given inorder and preorder traversal arrays. Display the tree using level order traversal.
6. Write a program to create a Binary Search Tree and implement insert() and search() operations.
7. Extend the previous BST program to support deletion of nodes (handling all three cases: leaf, one child, two children). Display in-order after deletion.
8. Write a program to create a Threaded Binary Tree and perform in-order traversal without recursion or stack.
9. Extend the threaded binary tree implementation to support preorder traversal using threading logic.
10. Write a program to create a Min-Heap and Max-Heap using an array. Include operations to insert(), delete(), and heapify() the structure.
11. Write a program to sort an array using Heap Sort. Display the original and sorted array using both Min-Heap and Max-Heap logic.

	Total Lecture Hours	**105 hours**

Textbook:

1. Horowitz, E., Sartaj Sahni, & Anderson-Freed, S. (2008). Fundamentals of data structures in C. University Press.
2. Lipschutz, S. (2014). Data structures. Mcgraw Hill Education (India) Private Limited.
3. Deshpande, P. S., & Kakde, O. G. (2009). C and data structures. Dreamtech Press.
4. Aaron M. Tenenbaum, Langsam, Y., & Augenstein, M. (2003). Data Structures Using C.

Reference Books:

1. Aho, A. V., Hopcroft, J. E., & Ullman, J. D. (2009). Data Structures and algorithms. Dorling Kindersly.
2. Kruse. (n.d.). Data Structures and Program Design in C. Pearson Education India.
3. Kernighan, B. W., & Ritchie, D. M. (2015). The C programming language. Pearson.
4. Van, P. (1994). Expert C programming: deep C secrets. Sunsoft Press.
5. Deitel, P., & Deitel, H. (2016). C How to Program, Global Edition. Pearson Higher Ed.

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
100	25	**125**	**250**
125			

Course Code: 26CS102ESB	**Course Name: Computer Organization and Architecture**	**L**	**T**	**P**	**C**
		2	**0**	**2**	**3**

Pre-requisite: Basic knowledge of digital logic design, computer fundamentals, and programming concepts.

Co-requisite: NA

Anti-requisite: NA

Course Objectives:

1. To provide a foundation in data representation, instruction formats, and basic operations of a computer system.
2. To introduce the principles of processor design, pipelining, and performance evaluation.
3. To develop understanding of memory organization, cache mechanisms, and storage hierarchy.
4. To familiarize students with advanced computer architectures, including parallel and multiprocessor systems.

Course Outcome: After completion of the course, the student will be able to

1. Apply data representation techniques, micro-operations, and arithmetic algorithms in computer systems.
2. Examine instruction set architecture and processor control unit operations.
3. Analyse the influence of pipelining and hazard handling on processor performance.
4. Analyse memory hierarchy and cache mechanisms to enhance system performance.
5. Classify parallel architectures and multiprocessor systems based on their organization and operation.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	-	1	-	-	-	-	-	1
CO2	3	2	1	-	1	-		-	-	-	1
CO3	3	3	1	1	1	-	-	-	-	-	1
CO4	3	3	1	1	1	-	-	-	-	-	1
CO5	3	2	1	-	1	-	-	-	-	-	2

Unit 1 Fundamentals of Computer Organization — **12 hours**

Data Representation: Fixed-point representation, Floating-point representation, IEEE 754 floating point standard, Signed number representation: 1's complement, 2's complement, Overflow and underflow, Micro-Operations: Arithmetic operations, Logic operations, Shift operations, Arithmetic Algorithms: Multiplication (Booth's Algorithm), Division (Restoring and Non-Restoring methods)

Indicative Program:

- Design and simulation of register transfer and micro-operations.

Unit 2 Instruction Set Architecture and Processor Control — **12 hours**

Addressing modes: Immediate, Direct, Indirect, Indexed, etc., Instruction format and types, Instruction execution cycle (Fetch–Decode–Execute), Control Unit Organization: Hardwired and Microprogrammed Control

Indicative Program:

- Implementation of different addressing modes instructions in 8085 microprocessors.

Unit 3 Processor Design and Performance — **12 hours**

Basic Processor Organization, Instruction Pipelining: Pipeline concept and stages, Pipeline Hazards: Data hazards, Control hazards, Structural hazards, Hazard handling techniques, Performance Evaluation: Speedup, Throughput, CPI (Cycles Per Instruction), Advanced Processor Architectures: Superscalar architecture, VLIW architecture.

Indicative Program:

- Simulation of 4 stage instruction pipeline.

Unit 4 Memory Organization — **12 hours**

Memory Hierarchy, Cache Memory: Concept and organization, Mapping techniques, Locality of Reference, Replacement policies, Advanced Cache Concepts: Multilevel caches, Caches in multi-core processors, Main Memory: RAM and ROM (Organization, Operation, Applications), Secondary Memory: Disk storage, Solid State Drives (SSD), Virtual Memory

Indicative Program:

- Memory Hierarchy Demonstration such that Cache Hit LED ON → Data from Cache , Cache Miss → RAM Access LED ON, RAM Miss → Disk Access LED ON

Unit 5 Advanced Computer Architecture — **12 hours**

Classification of Parallel Systems: SISD, SIMD, MIMD, SPMD, Vector processors, Multiprocessor Systems: Shared memory systems, Distributed memory systems, Cluster computing and message passing systems, Memory Issues in Multiprocessors: Memory consistency models, Cache coherence, Interconnection Networks, Introduction to GPU, Performance Evaluation: Multiprocessor benchmarks.

Indicative Programs:

- Simulation of SIMD (Single Instruction Multiple Data) Architecture using Parallel ALUs.

Total Lecture Hours — **60 hours**

Textbook:

1. Computer Organization and Design: The Hardware/Software Interface, David A. Patterson and John L. Hennessy, 5th Edition, Morgan Kaufmann.
2. Computer Organization and Architecture: Designing for Performance, William Stallings, 10th Edition, Pearson Education.
3. Computer System Architecture, M. Morris Mano, 3rd Edition, Pearson Education.

Reference Books:

1. Computer Architecture: A Quantitative Approach, John L. Hennessy and David A. Patterson, 6th Edition, Morgan Kaufmann.
2. Advanced Computer Architecture: Parallelism, Scalability, Programmability, Kai Hwang, McGraw-Hill Education.
3. Modern Processor Design: Fundamentals of Superscalar Processors, John P. Shen and Mikko H. Lipasti, McGraw-Hill Education.

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
60	15	**75**	**150**
75			

Practical Courses Detail Syllabus

Course Code: 26AS101HSP	Course Name: Communication Skills	L	T	P	C
		0	0	2	1

Pre-requisite: Basic proficiency in English at the school level (as per qualifying examination standards)

Co-requisite: Concurrent engagement with adaptive learning modules and digital platforms (such as WeXL and FluentEdge) prescribed for the course.

Anti-requisite: NA

Course Objectives:

1. Develop a foundational understanding of communication, including voice dynamics for effective speech delivery.
2. Enhance reading comprehension, note-making, and idea organization skills for academic and professional use.
3. Strengthen writing proficiency through vocabulary building and application of effective writing techniques.
4. Build essential listening and professional communication skills for employability.

Course Outcome: After completion of the course, the student will be able to

1. Analyze and utilize feedback from adaptive learning platforms (such as WeXL) to regulate and improve individual performance across LSRW domains.
2. Apply listening, speaking, reading, and writing strategies to produce clear, accurate, and contextually appropriate communication in structured tasks.
3. Construct coherent and organized messages in digital and interactive communication settings, demonstrating clarity, language control, and audience awareness.
4. Analyze communication situations and respond through logical reasoning, active listening and collaborative engagement in group contexts.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	-	2	3	-	2
CO2	-	-	-	-	-	-	-	2	3	-	2
CO3	-	-	-	-	-	-	-	2	3	-	2
CO4	-	-	-	-	-	-	-	2	3	-	2

Unit-1	Fundamentals of Communication and LSRW Skills	8 hours

•Nature and process of communication
•Types and barriers to communication
•Introduction to Listening, Speaking, Reading, and Writing (LSRW)
•Diagnostic assessment through WeXL (BET)
•Orientation to Fluent Edge modules.

Unit 2	Self-Expression and Digital Communication	8 hours

•Structuring ideas effectively (organization and coherence)
•Tone, register, and appropriate language use
•Communication in digital contexts
•Fundamentals of video-based communication: voice modulation, kinesics, confidence

Unit 3	Interactive Communication and Argumentation	7 hours

•Principles of group communication
•Active listening and response strategies
•Introduction to argumentation and reasoning
•Role-based communication in structured contexts

Unit 4	Applied Communication and Critical Reflection	7 hours

•Communication in problem-solving and decision-making situations
•Analytical discussion and reflective communication
•Responding to complex communication scenarios

Hybrid Learning Component
Adaptive learning through WeXL

Total Hours	30 hours

Mode of Evaluation

Evaluation Scheme			
MSE	CA	ESE	Total Marks
-	25	25	50
25			

Course Code: 26PH103BSP	Course Name: Engineering Physics Lab	L	T	P	C
		0	0	2	1

Pre-requisite: Basic understanding of Physics concepts, mathematical skills, familiarity with measurement techniques, error analysis, graph plotting, and safe handling of laboratory instruments and experimental procedures.

Co-requisite: NA

Anti-requisite: NA

Course Objectives:

To impart the experimental knowledge of semiconductor Physics and devices to engineering graduates so that they are able to assess and contribute to the solution of technical and engineering problems that are based on broad principles of Physics including Quantum Physics, Semiconductors, Diodes, Bipolar Junction Transistors and Optoelectronics devices.

Course Outcome: After completion of the course, the student will be able to

1. Apply Quantum physics principles to investigate discrete atomic energy levels using the Franck–Hertz experiment and determine Planck's constant and work function using the photoelectric effect.
2. Apply semiconductor device principles to analyse the characteristics of NPN/PNP transistors, MOSFETs, and solar cells, including maximum power point and fill factor.
3. Apply laser and optical-fibre principles to determine the wavelength of laser light, analyse the interference pattern produced by an optical fibre, and evaluate its attenuation and numerical aperture.
4. Apply magnetic material principles and experimental techniques to determine the magnetic susceptibility of a paramagnetic substance using Quincke's method.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	2	-	-	-	1	2	-	1
CO2	3	3	-	2	-	-	1	2	2	-	1
CO3	3	2	-	2	-	-	1	2	2	-	1
CO4	3	2	-	2	-	-	1	2	2	-	1

Experiment No.

1. To study the presence of discrete energy levels in an atom by Franck Hertz experiment.
2. To determine Planck's constant and work function using Photo-electric effect.
3. To study the characteristics of NPN/PNP transistors.
4. To study the V-I characteristics of MOSFET.
5. To study the V-I characteristics and power output characteristics of the solar cell and show the maximum power point on graph and calculate fill factor.
6. To determine the wavelength of Laser light using diffraction phenomena.
7. To determine the interference pattern produced by an optical fiber.
8. To find the fiber attenuation and numerical aperture of a given optical fiber.
9. Determination of Magnetic Susceptibility of a Paramagnetic Substance Using Quincke's Method
10. Measurement of Dielectric Constant Using Parallel Plate Capacitor

Total Hours	30 hours

Mode of Evaluation

Evaluation Scheme			
MSE	CA	ESE	Total Marks
-	25	25	50
25			

Course Code: 26AS102MCP	Course Name: Basic Proficiency in Japanese	L	T	P	C
		0	0	4	NC

Pre-requisite: Basic Knowledge of grammar and parts of speech.

Co-requisite: Adaptability towards foreign cultures.

Anti-requisite: NA

Course Objectives:

1. To develop basic listening, speaking, reading and writing proficiency in the target language, enabling global communication skills.
2. To foster an understanding of intercultural communication, particularly between India and target-language- speaking cultures.
3. Introduce students to the global significance of the target language and appreciate its culture & ethics in international contexts.

Course Outcome: After completion of the course, the student will be able to

1. Introduce themselves in the respective language and understand the syllables and number
2. Understand how language and culture interact in global context and impact intercultural communication
3. Apply their learning in basic conversations and understand the social etiquette of professional world
4. Utilize the skills of listening, speaking and non-verbal communication in the target language

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	-	1	3	-	2
CO2	-	-	-	-	-	-	-	2	3	-	2
CO3	-	-	-	-	-	-	-	3	3	-	2
CO4	-	-	-	-	-	-	-	3	3	-	2

Unit-1	Basic Sentence Structure	15 hours

- Basic Declarative and Interrogative Sentences
- Demonstrative Pronouns and Adverbs of Place
- Numbers, Family Members, Pronouns, Places, Countries and Occupations
- Counter Suffix: Sai 歳 ,Kai 階
- Particles: Wa は , Ka か , Mo も , No の

Unit 2	Basic Tenses and Verbs	15 hours

- Non-Past and Past Tenses
- Transitive and Intransitive Verbs
- Verbs of Displacement
- Time, Food, Vehicles, Places
- Counter Suffix: Ji 時 , Fun 分 , Gatsu 月 , Nichi 日 , Nen 年 , Kagetsu ヶ月 , Kan 間
- Particles: E へ , Ni に , O を , De で , Ga が

Unit 3	Adjectives and Qualities	15 hours

- I- Type and Na- Type adjectives
- Verbs of Transaction
- Adjective and Noun Conjunction
- Request and Asking
- Counter Suffix: Tsu つ , Mai 枚 , Dai 台 , Hon 本
- Particles: Ni に

Unit 4	Complex Verb Conjugation	15 hours

- Te- Form, Nai- Form, Tai- Form and their uses
- Desires
- Permission, Leniency, Prohibition and Compulsion
- Like, Dislike, Expertise and Ineptitude
- Particles: Ga が

	Total Hours	60 hours

Mode of Evaluation

Evaluation Scheme			
MSE	CA	ESE	Total Marks
-	50	-	50
50			

Course Code: 26AS111MCP	Course Name: Basic Proficiency in German	L	T	P	C
		0	0	4	NC

Pre-requisite: Basic Knowledge of grammar and parts of speech.

Co-requisite: Adaptability towards Foreign Culture and willing to learn nuances of Foreign Land

Anti-requisite: NA

Course Objectives:

1. To develop basic listening, speaking, reading and writing proficiency in the target language, enabling global communication skills.
2. To foster an understanding of intercultural communication, particularly between Indian and target-language- speaking cultures.
3. Introduce students to the global significance of the target language and appreciate its culture and ethics in international contexts.

Course Outcome: After completion of the course, the student will be able to

1. To Understand the basic parts of speech of the language, put up basic questions and be able to introduce themselves in respective language and understand the syllables and numbers.
2. Understand how language and culture interact in global context and impact intercultural communication.
3. Apply their learning in basic conversations, sentence building and understand the social etiquette of professional world
4. Utilize the skills of listening, speaking and non-verbal communication in the target language.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	-	1	3	-	2

CO2	-	-	-	-	-	-	-	2	3	-	2
CO3	-	-	-	-	-	-	-	3	3	-	2
CO4	-	-	-	-	-	-	-	3	3	-	2

Unit-1	**Introduction to Basic Components of German Language**	**15 hours**

Die Begrüßungen (Basic Greetings), Sich Vorstellen (Self-Introduction), Das Alphabet und Die Zahlen (Alphabet and Number System)
Grammar- Die Regelmäßige und Unregelmäßige Verben- Introduction to Basic Verbs
Grammar- Die Frage Bilden und Imperativ- Question Making and understand day to day life instructions

Unit 2	**Navigating Everyday Situation with Basic Vocabulary**	**15 hours**

Grammar - Das Konzept von Artikeln, Adjektive und Die Negation - Introduction to New Vocabulary - Topics from. Everyday walks of Life and Negations, Giving Directions using simple prepositions and navigating directions.
Grammar - Die Uhrzeit und Das Datum - Concept of Date and Time
Grammar - Die Modal Verben und Die Trennbare Verben – The Modal Verbs and Separable Verbs

Unit 3	**Communication in German**	**15 hours**

Grammar – Die Possessiv Pronomen - Introduction to Possessive Pronouns
Grammar - Introduction of basic sentence Making: Nomminativ, Akkusativ, dativ
Grammar- , Wechsel Präpositionen (Changeable Prepositions)

Unit 4	**Understanding Indian and German Cultures**	**15 hours**

Introduction to Vocabulary: Festival, Celebrations
Grammar: Partizip Perfekt (Present Perfect Tense)
Professional and non-professional communication: E-mail Writing and Form Filling Exercises

	Total Hours	**60 hours**

Mode of Evaluation

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
-	50	**-**	**50**
50			

Course Code: 26AS112MCP	**Course Name: Basic Proficiency in French**	**L**	**T**	**P**	**C**
		0	**0**	**4**	**NC**

Pre-requisite: Basic Knowledge of grammar and parts of speech.

Co-requisite: Adaptability towards Foreign Culture and willing to learn nuances of Foreign Land

Anti-requisite: NA

Course Objectives:

1. To develop basic listening, speaking, reading and writing proficiency in the target language, enabling global communication skills.
2. To foster an understanding of intercultural communication, particularly between Indian and target language speaking cultures.
3. Introduce students to the global significance to the target language and appreciate its culture & ethics in international contexts.

Course Outcome: After completion of the course, the student will be able to

1. Introduce themselves in the respective language and understand the syllables and numbers
2. Apply their learning in basic conversation and understand the social etiquette of professional world
3. Understand how language and culture interact in global context and impact intercultural communication
4. Utilize the skills of listening, speaking and non-verbal communication in the target language

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**
CO1	-	-	-	-	-	-	-	2	3	-	2
CO2	-	-	-	-	-	-	-	2	3	-	2
CO3	-	-	-	-	-	-	-	1	2	-	2
CO4	-	-	-	-	-	-	-	3	2	-	3

Unit-1	**Basic Component of French Language**	**15 hours**

- Se saluer (Express and Understand basic greetings)
- L'alphabet (Alphabets)
- Les accents (Accents & Symbols)
- Les nombres (The Numbers)
- Les pronoms sujets (Subject pronouns)
- Le présent (régulier/irrégulier)
- Les phrases négatives (negative sentences)
- Décrire une personne (Describe a person)
- Les nationalités/métiers (nationalities/professions)

Unit 2	**Expressing Needs and Asking Questions**	**15 hours**

- Les articles (définis/indéfinis)

- Les objets de la classe et de la maison (Vocabulary of the class and house)
- Les adjectifs possessifs (The possessive adjectives)
- Les questions fermées (The closed questions)
- Les adjectifs interrogatifs (The interrogative adjectives)
- Cultural Focus: French social etiquettes, mealtime behaviour

Unit 3	**Basic Communication and Social Etiquettes**	**15 hours**

- La Famille (The Family)
- Parler de sa famille (Speak about one's family)
- Le Futur Proche (Near future tense)
- Le présent progressif (Present continuous)
- Le passé récent (recent past tense)
- L'heure (The time)
- Cultural Focus: Formal and Informal communication in French

Unit 4	**Respecting French and Indian Cultures in Global Context**	**15 hours**

- Les verbes pronominaux (The pronominal verbs)
- Raconter sa journée quotidienne (Tell about one's daily routine)
- La Carte Postale
- La fiche d'inscription (Information Form)
- Les loisirs (The hobbies)
- Compréhension écrite (Reading)
- Parler en classe de votre vie quotidienne en parlant de vos loisirs (Speak about your daily routine while talking about your hobbies)

	Total Hours	**60 hours**

Mode of Evaluation

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
-	50	-	**50**
50			

Course Code: 26AS113MCP	**Course Name: Basic Proficiency in Spanish**	**L**	**T**	**P**	**C**
		0	**0**	**4**	**NC**

Pre-requisite: Basic Knowledge of grammar and parts of speech.

Co-requisite: Adaptability towards Foreign Culture and willing to learn nuances of Foreign Land

Anti-requisite: NA

Course Objectives:

1. To develop basic listening, speaking, reading and writing proficiency in the target language, enabling global communication skills.
2. To foster an understanding of intercultural communication, particularly between India and target-language- speaking cultures.
3. Introduce students to the global significance of the target language and appreciate its culture & ethics in international contexts.

Course Outcome: After completion of the course, the student will be able to

1. Introduce themselves in the respective language and understand the syllables and number.
2. Understand how language and culture interact in global context and impact intercultural communication.
3. Apply their learning in basic conversations and understand the social etiquette of professional world.
4. Utilize the skills of listening, speaking and non-verbal communication in the target language.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**
CO1	-	-	-	-	-	-	-	2	3	-	2
CO2	-	-	-	-	-	-	-	2	3	-	2
CO3	-	-	-	-	-	-	-	1	2	-	2
CO4	-	-	-	-	-	-	-	3	2	-	3

Unit-1	**Basics of Communication & Identity**	**15 hours**

(En el aula + Nosotros y nosotras + Quiero aprender español)

- Greetings and classroom expressions
- Alphabet and pronunciation
- Introducing oneself and others
- Nationalities and professions
- Talking about reasons for learning Spanish

Grammar:

- Subject pronouns
- Verb *ser*
- Regular verbs (present tense)

- Interrogatives (qué, cómo, quién)
- Connectors (y, pero, porque)

Communicative Functions:

- Giving personal information
- Asking and answering basic questions

Unit 2	Location, Preferences & Social Interaction	15 hours

(¿Dónde está Santiago? + ¿Cuál prefieres? + Tus amigos son mis amigos)

- Talking about countries, cities, and location
- Asking/giving directions
- Expressing likes, dislikes, and preferences
- Talking about friends and relationships

Grammar:

- Verb *estar* and *hay*
- Prepositions of place
- Verb *gustar*
- Demonstratives (este, ese, aquel)
- Possessives (mi, tu, su)
- Verb *tener*

Communicative Functions:

- Describing people and relationships
- Expressing preferences

Unit 3	Daily Life & Food	15 hours

(Día a día + ¡A comer!)

- Daily routine and habits
- Time, days, schedules
- Food and drink vocabulary
- Ordering food

Grammar:

- Reflexive verbs
- Verb *ir* (voy a…)
- Quantifiers (mucho, poco)
- Present tense usage

Communicative Functions:

- Talking about routine
- Ordering food and expressing quantity

Unit 4	Places & Descriptions	15 hours

(El barrio ideal + overall revision)

- Describing neighborhoods and cities
- Talking about surroundings
- Basic cultural aspects

Grammar:

- Adjectives (agreement and usage)
- Prepositions of place
- Revision of *ser vs estar*

Communicative Functions:

- Describing places
- Giving simple descriptions

	Total Hours	60 hours

Mode of Evaluation

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
-	50	-	**50**
50			

Course Code: 26AS103MCP	Course Name: Indian Knowledge System	L	T	P	C
		0	0	2	NC

Indian Knowledge System: Students need to choose any one from the Courses offered

1. Case Study of Indian water storage system
2. Case study of Indian urban planning (Indus valley civilization)

3. Learning of Geeta for Engineers
4. Vasudhaiva kutumbakam: Indian model of multiculturalism
5. Impact of Satvik Food on the Gut-Microbiome Diversity
6. Review Socialism in Light of Ramayana (Critical Thinking)
7. The Relevance and Applicability of Chanakya's (Kautilya) Arthaśāstra for Solving Current Societal Problems
8. Lessons of Leadership from Mahabharat
9. Study of Ancient Indians Technology for Extraction, Purification, and Alloying of Metals such as Gold, Silver, Copper, and Iron
10. Study of Herbs used in Kitchen for Healthy Life(Haldi, Garlic etc.)
11. Study of Indian Astrology System
12. Study Significance of the Asanas , Pranayams and Surya Namaskar
13. Study of Vastu Shastra
14. Study of Ayurveda
15. Orientation of Temples of South India and their Astronomical Associations
16. Corporate Social Responsibility: A Philosophical Social Engineering Approach from an Ancient Indian Perspectives.

*The topics listed above are indicative and not exhaustive. Students may opt for any other relevant IKS topic, subject to approval by the concerned faculty mentor and the competent authority.

Total Hours: 30 hrs.

Mode of Evaluation

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
-	50	-	**50**
50			

Course Code: 26AS201HSP	**Course Name: Public Speaking Skills**	**L**	**T**	**P**	**C**
		0	**0**	**2**	**1**

Pre-requisite: Successful completion of Communication Skills (Semester I) or equivalent foundational training in communication.

Co-requisite: Concurrent engagement with adaptive learning platforms (such as WeXL and FluentEdge) and participation in performance-based classroom activities.

Anti-requisite: NA

Course Objectives:

1. To develop effective public speaking skills through structured delivery, voice modulation, and appropriate non-verbal communication.
2. To enable students to organize and present ideas using visual, verbal, and narrative strategies suited to diverse communication contexts.
3. To cultivate persuasive and analytical communication through argumentation, discussion, and debate.
4. To enhance adaptive communication abilities through performance-based activities and feedback-driven learning.

Course Outcome: After completion of the course, the student will be able to

1. Analyze and utilize feedback from adaptive learning platforms (such as WeXL) to regulate and improve individual performance across LSRW domains.
2. Apply listening, speaking, reading, and writing strategies to produce clear, accurate, and contextually appropriate communication in structured tasks.
3. Construct coherent and organized messages in digital and interactive communication settings, demonstrating clarity, language control, and audience awareness.
4. Analyze communication situations and respond through logical reasoning, active listening and collaborative engagement in group contexts.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	-	2	3	-	2
CO2	-	-	-	-	-	-	-	2	3	-	2
CO3	-	-	-	-	-	-	-	2	3	-	2
CO4	-	-	-	-	-	-	-	2	3	-	2

Unit-1	**Principles of Public Speaking and Performance**	**8 hours**

- Elements of effective public speaking
- Voice modulation, articulation, and pacing
- Body language (kinesics) and stage presence
- Audience awareness and engagement strategies
- Adaptive skill development through WeXL

Unit 2	**Presentation Design and Visual Communication**	**8 hours**

- Organization of speech (introduction, body, conclusion)
- Clarity, coherence, and audience alignment

- Use of visual aids and storytelling techniques
- Integrating personal voice with structured content

Unit 3	Collaborative Communication and Creative Expression	7 hours

- Team communication and coordination
- Role allocation and collaborative speaking
- Structuring multi-speaker presentations
- Creative and critical thinking in communication

Unit 4	Persuasive Communication and Debate	7 hours

- Structuring arguments and counterarguments
- Rebuttal techniques and critical questioning
- Audience engagement in competitive speaking contexts

Total Hours	30 hours

Mode of Evaluation

Evaluation Scheme			
MSE	CA	ESE	Total Marks
-	25	25	50
25			

Course Code: 26CT101PWP	Course Name: Design Thinking for Innovation and Entrepreneurship	L	T	P	C
		0	0	2	1

Pre-requisite: Basic understanding of problem-solving, creativity, and communication skills is desirable

Co-requisite: Participation in innovation, entrepreneurship, project-based learning, or skill development activities.

Anti-requisite: NA

Course Objectives:

1. Introduce students to the concepts of Design Thinking, innovation, entrepreneurship, and start-up ecosystems.
2. Develop empathy-driven problem identification and creative ideation skills for generating innovative solutions.
3. Familiarize students with market research, customer analysis, and business idea validation techniques.
4. Enable students to design prototypes/MVPs and effectively pitch innovative business ideas.

Course Outcome: After completion of the course, the student will be able to

1. Understand Design Thinking, innovation, and start-up ecosystems.
2. Apply empathy and ideation techniques to identify problems and generate solutions.
3. Analyze market needs and validate entrepreneurial opportunities.
4. Develop prototypes/MVPs and effectively pitch innovative business ideas.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	2	–	–	1	–	–	–	2	2
CO2	2	3	3	2	1	–	–	2	2	1	2
CO3	2	3	2	2	1	2	1	2	2	3	2
CO4	2	2	3	2	2	1	1	3	3	3	2

Unit-1	Introduction to Design Thinking, Innovation and Start-ups	8 hours

Introduction to Design Thinking phases, innovation concepts, creativity, types of innovation, start-up lifecycle, start-up ecosystem, and innovation case studies. Role of incubation centers, accelerators, and start-up ecosystems.

Unit 2	Empathy and Ideation	8 hours

Empathy mapping, customer journey map, personas, problem identification, brainstorming, mind mapping, idea generation methods, Introduction to Business Model Canvas (BMC).

Unit 3	Market Research and Entrepreneurship	7 hours

Market research, customer analysis, business idea validation, entrepreneurship basics and start-up success stories.

Unit 4	Prototype and Pitching	7 hours

Minimum Viable Product (MVP), prototype development and testing, feedback and improvement, start-up development, innovation pitching, and presentation skills.

Total Hours	30 hours

Mode of Evaluation

Evaluation Scheme			
MSE	CA	ESE	Total Marks
-	50	-	50
50			

Course Code: 26IT103PCP	Course Name: Web Designing and Cloud Foundations	L	T	P	C
		0	0	4	2

Pre-requisite: Basic knowledge of computer programming concepts is recommended for understanding web technologies and cloud computing concepts.

Co-requisite: NA

Anti-requisite: NA

Course Objectives:

This course aims to equip students with the knowledge and practical skills required to design and develop modern web applications and understand cloud computing fundamentals. It focuses on web technologies, web design and development, user interface design, content management systems, version control practices, cloud services and deployment models, DevOps concepts, REST APIs, containerization, and cloud security. The course enables learners to develop, deploy, manage, and host web and cloud-based applications using industry-standard tools and platforms.

Course Outcome: After completion of the course, the student will be able to

1. Understand web development fundamentals and apply HTML, HTML5, XML, and UI/UX design principles using Figma to build structured, accessible, and standards-compliant web interfaces.
2. Design responsive and interactive web applications using CSS, JavaScript, jQuery, and AJAX, and manage collaborative development and hosting using Git and GitHub.
3. Develop content-driven websites using Content Management Systems (WordPress) and apply cloud computing fundamentals, service and deployment models, and DevOps practices.
4. Implement REST API-based application workflows, containerize and orchestrate applications using Docker and Kubernetes, and apply IAM and cloud security mechanisms.
5. Develop, deploy, and manage scalable web and cloud-enabled applications using AWS and IBM Cloud platforms.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2	3	2	-	2	2	2	3
CO2	3	3	3	2	3	3	1	2	2	2	3
CO3	3	3	3	2	3	3	-	2	2	2	3
CO4	3	3	3	3	3	3	1	2	3	2	3
	3	3	3	3	3	3	1	2	3	2	3

Unit-1	Web Technology Fundamentals, Web Designing and HTML	12 hours

Introduction to Web Technologies and evolution of the World Wide Web • Understanding Internet architecture and website communication • Fundamentals of HTTP, HTTPS, URLs, DNS, and Web Browsers • Introduction to client-server computing architecture • Web project planning and website development workflow • HTML fundamentals: text formatting, lists, tables, images, frames, and forms • XML fundamentals: DTD, XML Schemas, and Object Models • Processing XML using DOM and SAX parsers • HTML5 semantic and non-semantic elements • HTML5 multimedia: audio and video integration • Introduction to UI/UX principles and wireframing using Figma • Creating website layouts and interface prototypes in Figma

Case Study: Design and develop a personal portfolio website using HTML5 and Figma. Create wireframes, incorporate semantic HTML elements, multimedia content, forms, and structured navigation to develop an accessible and standards-compliant website.

Industry Alignment: Knowledge of web technologies, HTML5, and UI/UX design forms the foundation of modern front-end development. Skills in browser architecture, HTTP communication, and prototyping using Figma are highly valued in roles such as Front-End Developer, UI Engineer, Web Designer, and Product Designer.

Unit 2	CSS, JavaScript, jQuery and Version Control with GitHub	12 hours

Creating and linking Cascading Style Sheets (CSS) • Working with block-level elements, lists, tables, CSS selectors, IDs, and classes • Advanced CSS concepts: positioning, layouts, floating, alignment, transitions, and pseudo-classes • Responsive web design principles and media queries • JavaScript fundamentals: variables, functions, forms, objects, and DOM manipulation • Event-driven programming and dynamic user interactions • Introduction to AJAX and asynchronous communication • Introduction to jQuery and implementation in web projects • Interactive components using Owl Carousel and LightBox plugins • Introduction to Version Control Systems (VCS) • Git fundamentals: repositories, branching, commit management • Collaborative development and project hosting using GitHub • Publishing and managing static websites through GitHub

Case Study: Develop a responsive company website with interactive UI components. Manage project versions using Git and publish project deliverables using GitHub workflows.

Industry Alignment: CSS, JavaScript, Git, and GitHub are core technologies used in modern software and web development. Version control and collaborative workflows are essential requirements for Front End Developers, Web Developers, and Full-Stack Engineers.

Unit 3	Content Management Systems, Cloud Foundations and DevOps	12 hours

Introduction to Content Management Systems (CMS) • WordPress fundamentals and website architecture • Installation and configuration of WordPress • Managing themes, plugins, menus, pages, and media • Designing dynamic websites using WordPress • Introduction to Cloud Computing: concepts, evolution, and architecture • Cloud Service Models: SaaS, PaaS, and IaaS • Cloud Deployment Models: Public, Private, and Hybrid Cloud • Introduction to DevOps principles and Continuous Delivery • DevOps lifecycle and collaboration practices • Fundamentals of REST architecture and API communication • CI/CD concepts and deployment workflows • Introduction to cloud-native application development

Case Study: Design and deploy a content-driven website using WordPress and explore deployment workflows using DevOps practices and

cloud deployment models.

Industry Alignment: Content Management Systems, cloud fundamentals, and DevOps practices are widely adopted across startups and enterprise environments. Knowledge of CMS platforms, deployment workflows, CI/CD, and cloud-native concepts prepares learners for careers in Web Development, Cloud Engineering, and Platform Operations.

Unit 4	**REST APIs, Containerization and Cloud Security**	**12 hours**

Principles and architecture of REST APIs • API request-response lifecycle and JSON data exchange • Integration with IBM Watson REST APIs • Docker fundamentals and container lifecycle • Container image creation and Docker Hub • IBM Cloud Container Registry and deployment concepts • Container orchestration using Kubernetes • Cloud-native application deployment fundamentals • Data protection and security in cloud environments • Identity and Access Management (IAM) concepts and implementation • Cloud database fundamentals and IBM Cloudant integration • Security best practices in distributed cloud systems

Case Study: Build a cloud-native application using REST APIs and containerize the deployment using Docker and Kubernetes while implementing IAM-based security and cloud database services.

Industry Alignment: REST APIs, Docker, Kubernetes, and cloud security are among the most sought-after skills in cloud computing and DevOps job roles. Kubernetes has become the industry standard for container orchestration, while API-driven architectures are the backbone of modern web and enterprise applications. Knowledge of IAM, cloud databases, and security practices is essential for roles such as Cloud Engineer, DevOps Engineer, Site Reliability Engineer (SRE), and Cloud Solutions Architect. IBM Cloud services further provide hands-on exposure to enterprise-grade cloud technologies used across industries.

Unit 5	**AWS and IBM Cloud Services**	**12 hours**

Amazon EC2 fundamentals and cloud infrastructure concepts • Amazon Machine Images (AMI) and instance management • Launching, configuring, and managing EC2 instances • Networking and storage for EC2 • Security Groups and Elastic IP management • Hosting web applications on EC2 instances • Amazon S3 fundamentals: buckets and objects • S3 storage classes and access management • Static website hosting using Amazon S3 • Versioning and lifecycle management • Introduction to IBM Cloud ecosystem and services • IBM Cloud Dashboard, Catalog, and Documentation • Compute and deployment services in IBM Cloud • IBM Cloud Object Storage fundamentals • Application deployment and hosting concepts in IBM Cloud • IBM Watson services and cognitive application overview • Security and access management in IBM Cloud • Comparative overview of AWS and IBM Cloud services

Case Study: Deploy a web application using AWS and explore application hosting and cloud service deployment using IBM Cloud. Configure infrastructure, storage, and security settings to ensure reliable and scalable application delivery.

Industry Alignment: AWS and IBM Cloud provide exposure to industry-standard cloud deployment and infrastructure management practices. Skills in virtual infrastructure, storage, deployment workflows, security, and cloud platforms prepare learners for cloud-focused roles and support pathways toward industry recognized certifications.

	Total Hours	**60 hours**

Text Books:

1. Xavier, C, "Web Technology and Design", New Age International
2. Burdman, Jessica, "Collaborative Web Development" Addison Wesley
3. Gautam Shroff, "Enterprise Cloud Computing Technology Architecture Applications", Cambridge University Press;2014.
4. IBM courseware.
5. Amazon Web Services AWS Cloud Practitioner Essentials

Reference Books:

1. Anubhav Hanjura, "Cloud Application Development",Packt Publishing Ltd, 2014.Peter Linz, "An Introduction to Formal Language and Automata", Narosa Publishing.
2. Toby Velte, Anthony Velte, Robert Elsenpeter, "Cloud Computing, A Practical Approach" McGraw-Hill Osborne Media; 2015

Mode of Evaluation

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
40	10	**50**	**100**
50			

Course Code: 26CE102PCP	**Course Name: Integrated Business Processes in Cloud ERP-I**	**L**	**T**	**P**	**C**
		0	**0**	**4**	**2**

Pre-requisite: Basic knowledge of business processes and computer fundamentals is recommended.

Co-requisite: NA

Anti-requisite: NA

Course Objectives:

This course introduces students to end-to-end business processes within SAP's Cloud ERP environment. It focuses on understanding how core enterprise functions such as Finance, Procurement, Sales, and Supply Chain are integrated within the SAP system landscape.

Course Outcome: After completion of the course, the student will be able to

1. Explain the SAP system landscape, SAP Cloud ERP architecture, and the role of SAP S/4HANA in integrating enterprise business processes.
2. Demonstrate proficiency in navigating the SAP Fiori interface and executing core business transactions within the SAP S/4HANA environment.
3. Execute and analyse integrated Procure-to-Pay (P2P) and Order-to-Cash (O2C) business processes, including the associated financial

postings in SAP.
4. Apply SAP best practices to perform and monitor integrated business scenarios across procurement, sales, finance, and supply chain functions.
5. Analyse, document, and evaluate end-to-end business process flows using SAP tools to support effective enterprise decision-making and process integration.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	2	3	2	1	2	2	2	3
CO2	3	3	3	2	3	2	1	2	2	2	3
CO3	3	3	3	3	3	3	1	2	3	2	3
CO4	3	3	3	3	3	3	1	2	3	2	3
CO5	3	3	3	2	3	3	-	2	2	2	3

Module I	**Introduction to SAP and Cloud ERP**	**20 hours**

- Overview of ERP and SAP S/4HANA Cloud
- SAP system landscape: Client, Mandant, and Organizational Structures
- Navigation in SAP Fiori Launchpad
- Key master data: Business Partner, Material, and Chart of Accounts
- Overview of SAP modules: FI, MM, SD, PP, HCM
- Introduction to integrated business processes
- SAP Best Practices and Solution Manager overview

Case Study: *Explore the SAP Fiori Launchpad and execute key navigation tasks across the integrated SAP S/4HANA Cloud environment.*

Industry Alignment: Understanding the SAP landscape is foundational for roles such as SAP Functional Consultant, Business Analyst, and ERP Implementation Specialist.

Module II	**Procure-to-Pay Process**	**20 hours**

- Purchase Requisition and Purchase Order creation
- Goods Receipt and Invoice Verification
- Vendor Master Data management
- Material Management integration with Finance
- Three-way match concept in procurement
- Accounts Payable postings and clearing

Case Study: *Execute a complete Procure-to-Pay cycle from purchase requisition to vendor payment in SAP S/4HANA Cloud.*

Industry Alignment: Procure-to-Pay skills are essential for SAP MM Consultants, Procurement Analysts, and Supply Chain professionals.

Module III	**Order-to-Cash Process**	**20 hours**

- Customer Master Data and Sales Org structures
- Sales Order creation and processing
- Delivery and Goods Issue process
- Billing and Invoice generation
- Accounts Receivable and Payment clearing
- Integration of SD with FI module

Case Study: *Process a full Order-to-Cash scenario from customer sales order creation through final payment receipt.*

Industry Alignment: Order-to-Cash process knowledge supports roles in SAP SD Consulting, Revenue Management, and Customer Service Operations.

	Total Hours	**60 hours**

Mode of Evaluation

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
40	10	**50**	**100**
50			